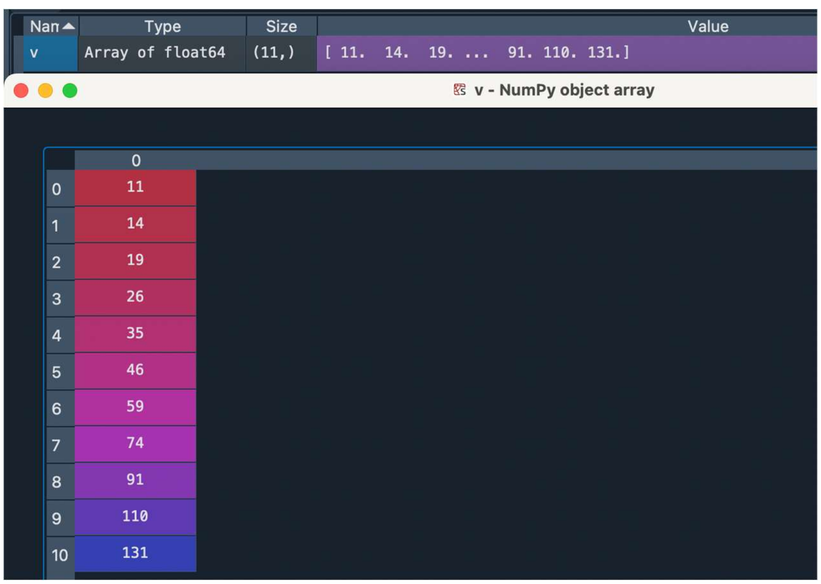
ES490 Homework 1 – Indexing and Solution of Linear Algebraic Equations

1. Indexing arrays and matrices is an important aspect of programming syntax to understand.
    - Note indexing follows the row x column standard. For example given a 2-d matrix A: A[row index, column index]. ***Be careful as Python indexing starts at zero!***
    - If you need to select more than 1 index, use colon ":" to specify an array of indices. For example b[0:3] would select the first 3 values in the b array. ***Note: the Python ending index stops before the value you want (i.e. 3 here, so Python will return b[0], b[1], and b[2] with this syntax) (which is really annoying) so you basically have to add 1 to your final index if you are selecting multiple elements in an array.***
    - If only a colon ":" is used (with no beginning and end indices) to index a dimension of an array, all indexes of that array are selected. For example A[:,0] would select all rows and the first column of matrix A.
    - Lastly, to index an array or matrix to the end of a dimension minus some number of indices you can use the indexing syntax [:-1]. For example b[:-1] selects all values of array b except for the last index. Or b[1:-1] selects all values of array b except for the first and last indexes.

Complete the following indexing exercises:

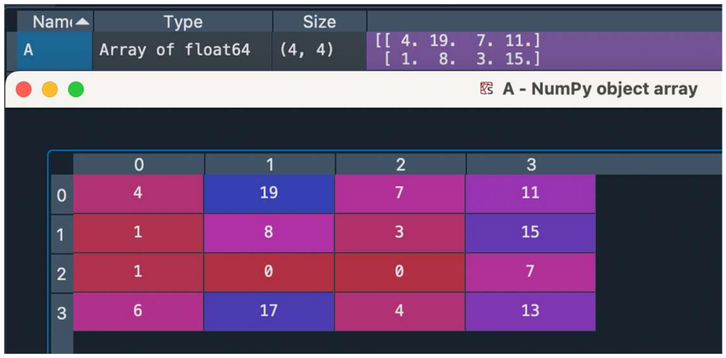
Given the Python variable window shown below:



a. Index the 10th element of v
   v[9]
b. Index the 4th through 7th elements of v
   v[3:6]

c. What values are returned if question b. was entered into the command window or run in a script?
[26, 35, 46, 59]

Given the Python variable window shown below:



d. Index the value of A in the 3rd row and 2nd column.
   A[2,1]
e. What value is returned if question d. was entered into the command window or run in a script?
   0
f. Index A to select the 2nd row and all columns of A.
   A[1,:]
g. What values are returned if question f. was entered into the command window or run in a script?

[1, 8, 3, 15]

h. Index A to select the 4th column and all rows except for the last row.

A[:-1, 4]

i. What values are returned if question h. was entered into the command window or run in a script?

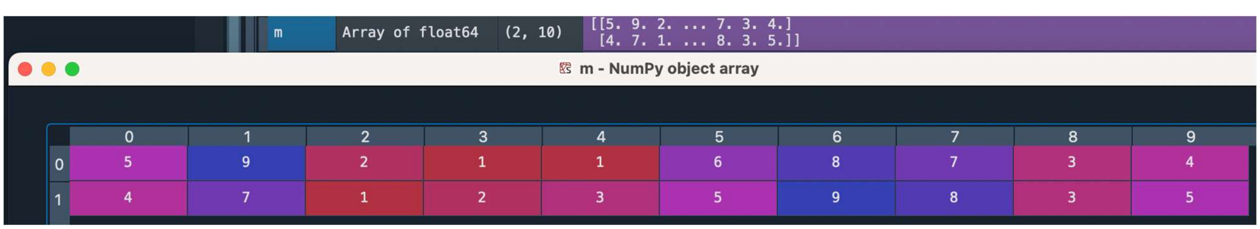
[11, 15, 7]

j. Lets say T is a 5 row x 50 column matrix preallocated with zeros (a must when creating and working with arrays and matrices with for loops in Python). This line of code for preallocation would read:

   import numpy as np
   T=np.zeros([5,50])

   Use two for loops to index every element of the matrix and set that element equal to i*j^2 by the time the for loops are completed. I have provided the correct double for loop syntax below. Fill in the correct indexing inside the for loops:

   for i in range(T.shape[1]): # iterate through the columns
     for j in range(T.shape[0]): # iterate through the rows
         T[i,j] = i*j^2;
k. You wish to calculate the average mass from 2 sets of measurements given in the first and second rows of variable m shown below:



While a mean function does exist in Python to accomplish this, please write 1 line of code manually (for indexing practice) in the given for loop to calculate the average mass for each trial in a variable called mavg:

```
import numpy as np
mavg=np.zeros(m.shape[1])

for i in range(m.shape[1]):

    mavg[i] = (m[0,i] + m[1,i])/2;
```

l. Write the code to calculate the average mass from part k using matrix notation instead (much faster for Python to compute when dealing with large arrays and matrices).

```
mavg = np.mean(m, axis=0);
```

2. A truss is loaded as shown in Figure 2.

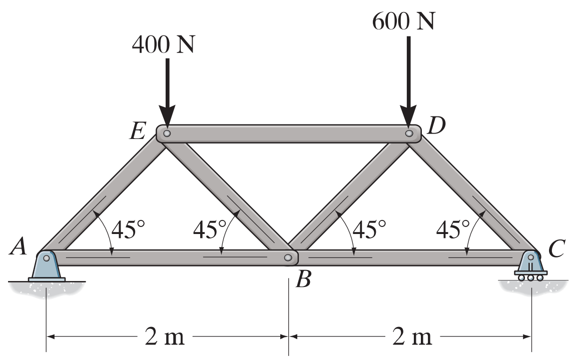


Figure 2: Truss loading

You are given the following sets of equations based on Method of Joints for joints A, B, and C.

$A_x + AB + AE*\cos(45) = 0$

$A_y + AE*\sin(45) = 0$

$-AB + BC - BE*\cos(45) + BD*\cos(45)=0$

$BE*\sin(45) + BD*\sin(45) = 0$

$-BC - CD*\cos(45) = 0$

$C_y + CD*\sin(45) = 0$

Assuming all members are in tension for sign consistency, write four additional equations using Method of Joints at joint D and E.

E: $+ED - EA\cos(45) + EB\cos 45 = 0$ $-400 - EA\sin(45) - EB\sin(45) = 0$	D: $-ED - DB\cos(45) + DC\cos(45) = 0$ $-600 - DB\sin(45) - DC\sin(45) = 0$

Solve for the 10 unknowns, AB, AE, BC, BD, BE, CD, DE, A_x, A_y, and C_y. For the sake of consistency (and my sanity for grading), lets all assume the same organization of our unknown vector:

$$x = \begin{Bmatrix} AB \\ AE \\ BC \\ BD \\ BE \\ CD \\ DE \\ Ax \\ Ay \\ Cy \end{Bmatrix}$$

[450., -636.39610307, 550., -70.71067812, 70.71067812, -777.81745931, -500., 0., 450., 550.]

2. A truss is loaded as shown in Figure 2.

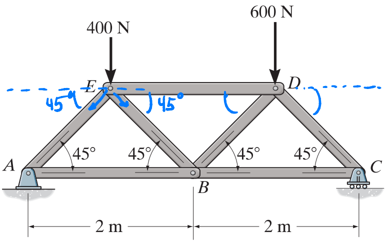


Figure 2: Truss loading

You are given the following sets of equations based on Method of Joints for joints A, B, and C.

$$Ax + AB + AE*\cos(45) = 0$$

$$Ay + AE*\sin(45) = 0$$

$$-AB + BC - BE*\cos(45) + BD*\cos(45) = 0$$

$$BE*\sin(45) + BD*\sin(45) = 0$$

$$-BC - CD*\cos(45) = 0$$

$$Cy + CD*\sin(45) = 0$$

$\Sigma F_{x@E} = +ED - EA\cos 45 + EB\cos 45 = 0$

$\Sigma F_{y@E} = -400 - EA\sin 45 - EB\sin 45 = 0$

D:

$\Sigma F_x = -ED - DB\cos 45 + DC\cos 45 = 0$

$\Sigma F_y = -600 - DB\sin 45 - DC\sin 45$

Assuming all members are in tension for sign consistency, write four additional equations using Method of Joints at joint D and E.

Solve for the 10 unknowns, *AB*, *AE, BC*, *BD, BE, CD, DE*, *Ax*, *Ay*, and *Cy*. For the sake of consistency (and my sanity for grading), lets all assume the same organization of our unknown vector:

3. A wall is constructed of concrete (grey) and insulation (blue) as shown in Figure 1. At equilibrium, the temperature profile (T_1, T_2, and T_3) in the wall is determined given T_0 and T_4 ambient air temperatures inside and outside the wall respectively. Heat transfer to the surfaces

$$
\begin{array}{r}
\\
\text{eq 1.} \\
\\
\\
\\
\\
\\
\\
\\
\\
\\
\end{array}
\begin{array}{c}
\begin{array}{cccccccccc}
AB & AE & BC & BD & BE & CD & DE & A_x & A_y & C_y
\end{array} \\
\begin{bmatrix}
1 & \cos 45 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & \sin 45 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
-1 & 0 & 1 & \cos 45 & -\cos 45 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \sin 45 & \sin 45 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & -\cos 45 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \sin 45 & 0 & 0 & 0 & 1 \\
0 & -\cos 45 & 0 & 0 & \cos 45 & 0 & 1 & 0 & 0 & 0 \\
0 & -\sin 45 & 0 & 0 & -\sin 45 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\cos 45 & 0 & \cos 45 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & -\sin 45 & 0 & -\sin 45 & 0 & 0 & 0 & 0
\end{bmatrix}
\end{array}
\begin{array}{c}
\dot{X} \\
\begin{bmatrix}
AB \\ AE \\ BC \\ BD \\ BE \\ CD \\ DE \\ A_x \\ A_y \\ C_y
\end{bmatrix}
\end{array}
=
\begin{bmatrix}
0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 400 \\ 0 \\ 600
\end{bmatrix}
$$

Verify solution:

@ D: $(-58.76) \quad (- \quad)$

$$\Sigma F_y = -600 - BD \sin(45) - CD \sin(45)$$

of the wall are governed by Newtons Law of Cooling: $q = h (T_f – T_i)$. Heat transfer through the wall materials are governed by Fouriers Law: $q = k/L (T_f – T_i)$.

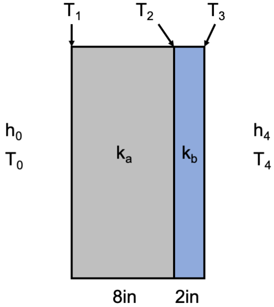


Figure 1: Composite slab subject to temperature boundary conditions

Write the 4 equations for heat transfer from the inside air into the concrete (q_1), through the concrete (q_2), through the insulation (q_3), and into the outside air (q_4):

$$q_1 = h_0(T_0 - T_1)$$
$$q_4 = h_4(T_3 - T_4)$$
$$q_2 = \frac{K_A}{8in}(T_1 - T_2)$$
$$q_3 = \frac{K_B}{2in}(T_2 - T_3)$$

To solve the system of equations, each heat transfer must equal according to conservation of energy $q_1=q_2$, $q_2=q_3$, and $q_3=q_4$. Express these 3 equations in matrix form:

$q_1 = q_2$
$$h_0(T_0 - T_1) = \frac{K_A}{8in}(T_1 - T_2)$$
$$h_0T_0 - h_0T_1 = \frac{K_A T_1}{8in} - \frac{K_A T_2}{8in}$$
$$\boxed{h_0T_0 - T_1\left(h_0 + \frac{K_A}{8in}\right) + T_2\frac{K_A}{8} = 0}$$

$q_2 = q_3$
$$\frac{K_A}{8}T_1 - \frac{K_A T_2}{8} = \frac{K_B}{2}T_2 - \frac{K_B}{2}T_3$$
$$\boxed{\frac{K_A}{8}T_1 + T_2\left(\frac{-K_A}{8} - \frac{K_B}{2}\right) + \frac{K_B}{2}T_3 \; (=0)}$$

$q_3 = q_4$
$$\frac{K_B}{2}T_2 - \frac{K_B}{2}T_3 = h_4T_3 - h_4T_4$$
$$T_2\left(\frac{K_B}{2}\right) + T_3\left(\frac{-K_B}{2} - h_4\right) + T_4(h_4)$$

$$\begin{array}{ccccc} T_0 & T_1 & T_2 & T_3 & T_4 \end{array}$$
$$\begin{bmatrix} h_0 & -\left(h_0 + \frac{K_A}{8}\right) & \frac{K_A}{8} & 0 & 0 \\ 0 & \frac{K_A}{8} & -\left(\frac{K_A}{8} + \frac{K_B}{2}\right) & \frac{K_B}{2} & 0 \\ 0 & 0 & \frac{K_B}{2} & -\left(\frac{K_B}{2} + h_4\right) & h_4 \end{bmatrix}$$

Solve for unknowns T_1, T_2, and T_3 given $T_0 = 70^{\circ}F$, $T_4 = 10^{\circ}F$, $h_0 = 2$ Btu/(hr-ft^2-$^{\circ}$F), $h_4 = 5$ Btu/(hr-ft^2-$^{\circ}$F), $k_a = 0.3$ Btu/(hr-ft-$^{\circ}$F), $k_b = 0.02$ Btu/(hr-ft-$^{\circ}$F):

Find: T_1, T_2, T_3

$T_0 = 70°F$
$T_4 = 10°F$

$h_0 = 2$ Btu/hr-ft^2-°F
$h_4 = 5$ Btu/hr-ft^2°F
$K_A = 0.3$ Btu/hr-ft°F
$K_B = 0.02$ " "

Solution:
Rearrange Matrix w/ givens

Solution:
Rearrange Matrix w/ givens

$T_0 = 70°F$
$T_4 = 10°F$

$h_0 = 2\ Btu/hr\text{-}ft^2\text{-}°F$
$h_4 = 5\ Btu/hr\text{-}ft^2\text{-}°F$
$K_A = 0.3\ Btu/hr\text{-}ft\text{-}°F$
$K_B = 0.02$ "

$$\begin{array}{ccccc} T_0 & T_1 & T_2 & T_3 & T_4 \end{array}$$
$$\begin{bmatrix} h_0 & -\left(h_0+\frac{K_A}{8}\right) & \frac{K_A}{8} & 0 & 0 \\ 0 & \frac{K_A}{8} & -\left(\frac{K_A}{8}+\frac{K_B}{2}\right) & \frac{K_B}{2} & 0 \\ 0 & 0 & \frac{K_B}{2} & -\left(\frac{K_B}{2}+h_4\right) & h_4 \end{bmatrix}$$

$$140\ Btu/hr\text{-}ft^2 + \begin{array}{c} \begin{array}{ccc} T_1 & T_2 & T_3 \end{array} \\ \begin{bmatrix} -\left(h_0+\frac{K_A}{8}\right) & \frac{K_A}{8} & 0 \\ \frac{K_A}{8} & -\left(\frac{K_A}{8}+\frac{K_B}{2}\right) & \frac{K_B}{2} \\ 0 & \frac{K_B}{2} & -\left(\frac{K_B}{2}+h_4\right) \end{bmatrix} \end{array} + 50\ Btu/hr\text{-}ft^2 = \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \rightarrow \begin{bmatrix} -\left(h_0+\frac{K_A}{8}\right) & \frac{K_A}{8} & 0 \\ \frac{K_A}{8} & -\left(\frac{K_A}{8}+\frac{K_B}{2}\right) & \frac{K_B}{2} \\ 0 & \frac{K_B}{2} & -\left(\frac{K_B}{2}+h_4\right) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} -140 \\ 0 \\ -50 \end{bmatrix} Btu/hr\text{-}ft^2$$

Now into Python!

Results:
$T_1 = 69.76°F$
$T_2 = 57.20°F$
$T_3 = 10.09°F$